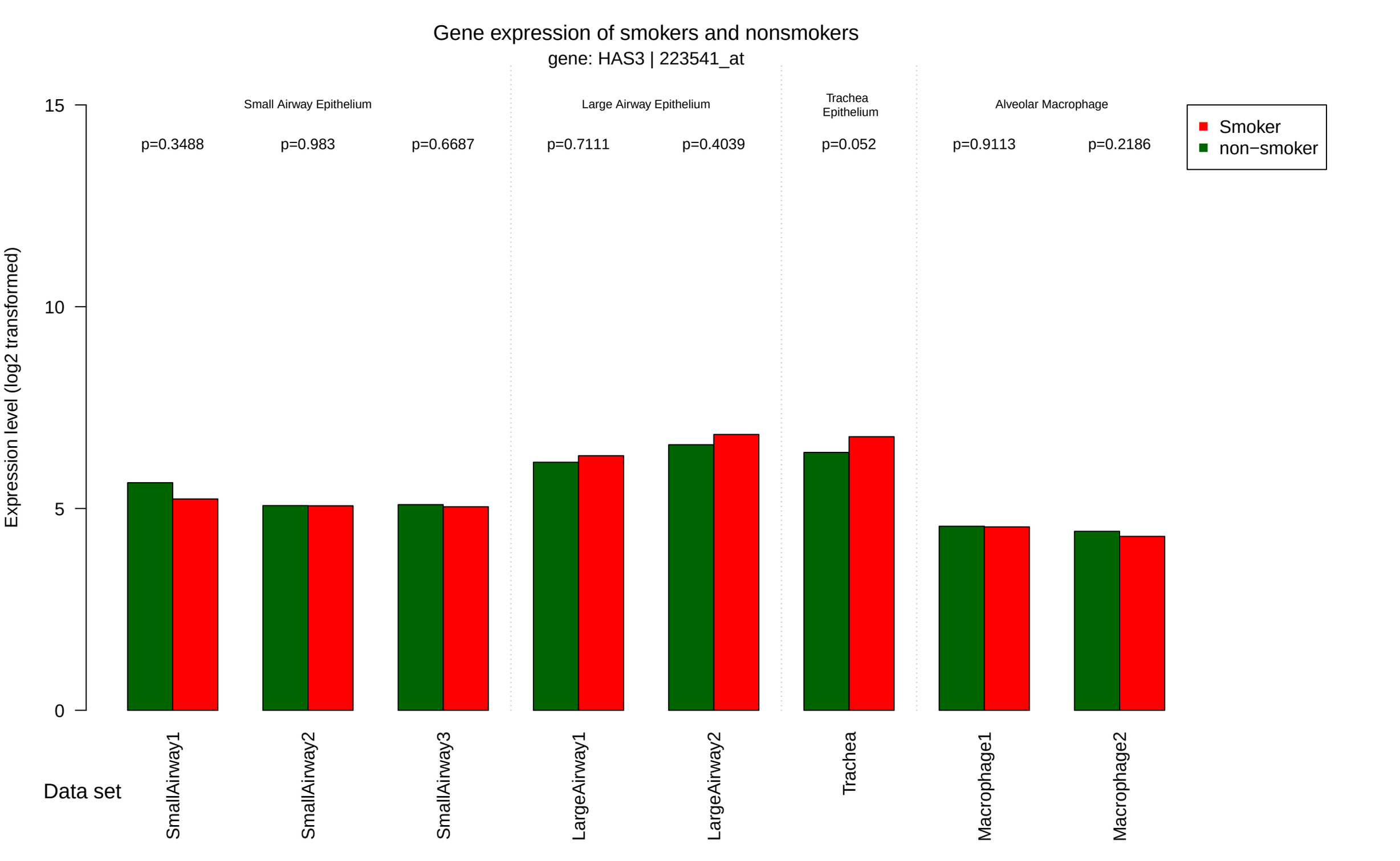
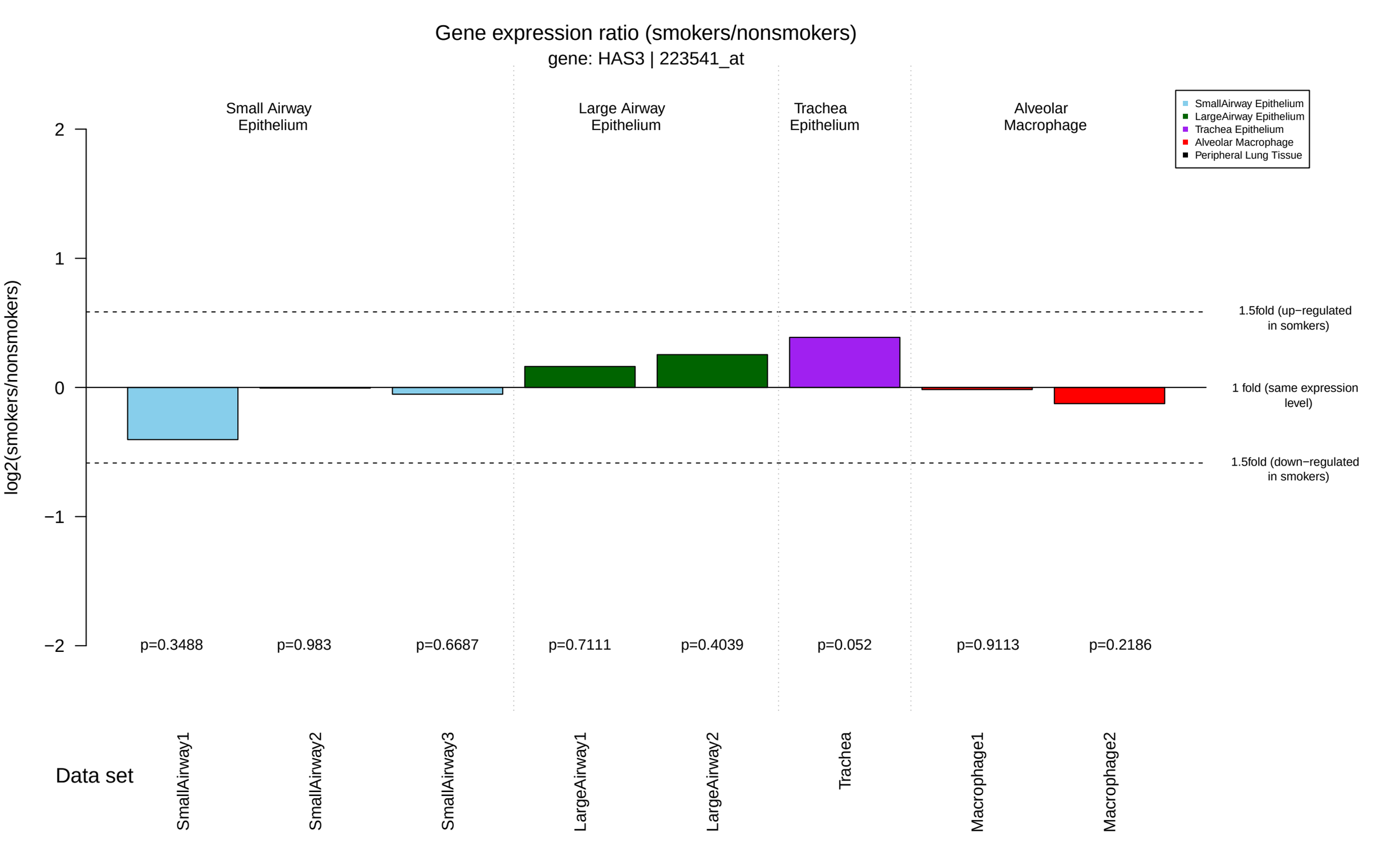
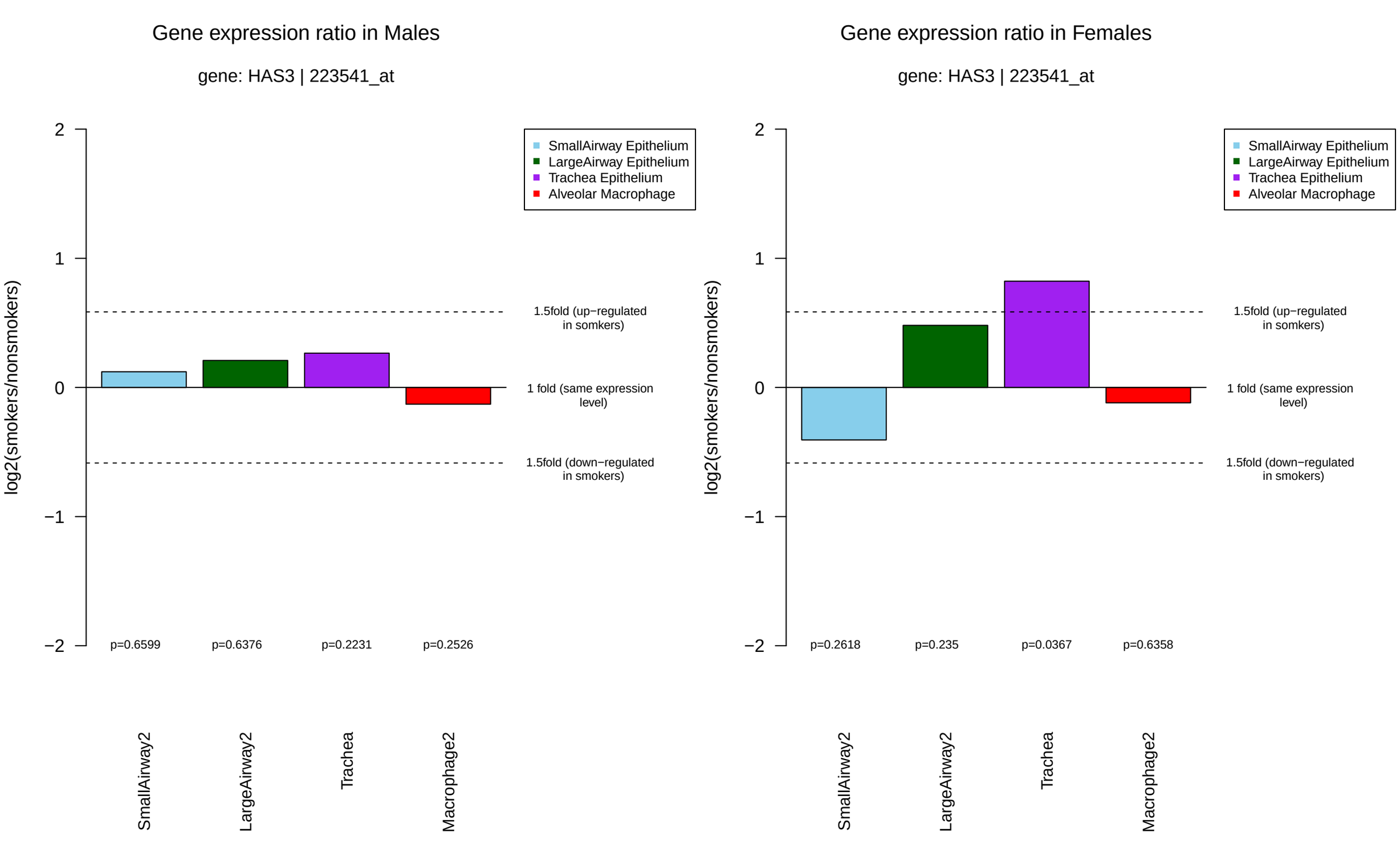
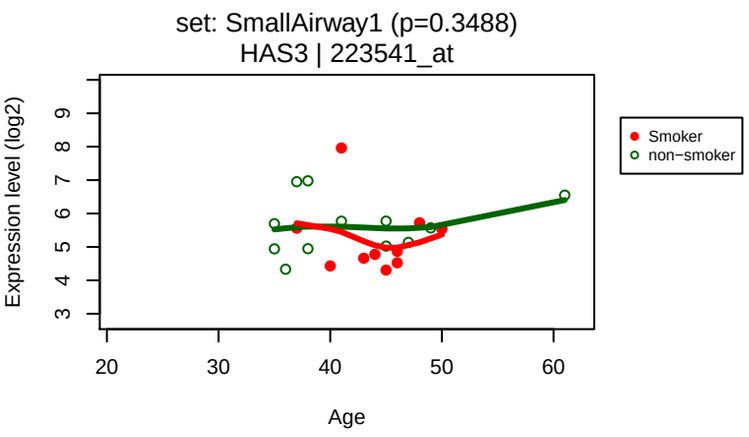
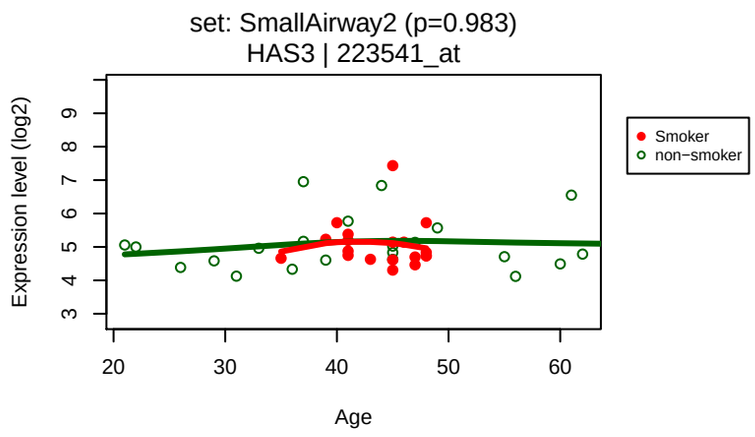
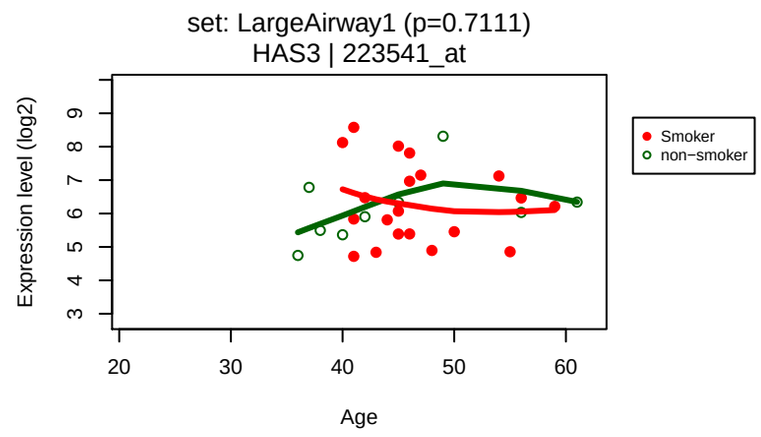

Gene expression of smokers and nonsmokers
gene: HAS3 | 223541_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3488
p=0.983
p=0.6687
p=0.7111
p=0.4039
p=0.052
p=0.9113
p=0.2186
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

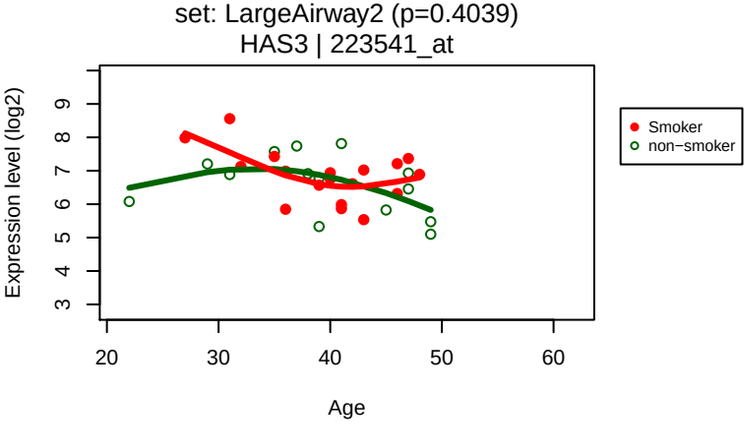

Gene expression ratio (smokers/nonsmokers)
gene: HAS3 | 223541_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3488
p=0.983
p=0.6687
p=0.7111
p=0.4039
p=0.052
p=0.9113
p=0.2186
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

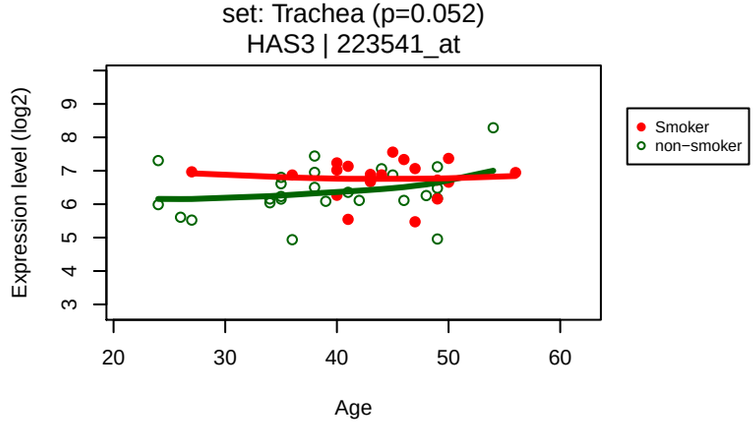

Gene expression ratio in Males
gene: HAS3 | 223541_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6599
p=0.6376
p=0.2231
p=0.2526
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HAS3 | 223541_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2618
p=0.235
p=0.0367
p=0.6358
SmallAirway2
LargeAirway2
Trachea
Macrophage2

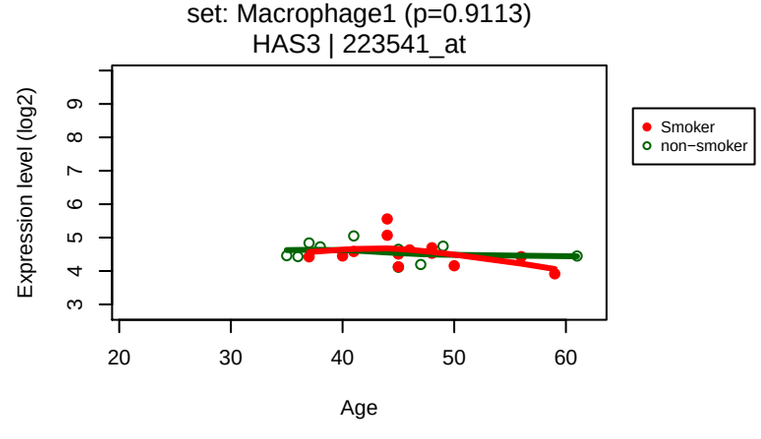

set: SmallAirway1 (p=0.3488)
HAS3 | 223541_at
Expression level (log2)
Age
Smoker
non−smoker

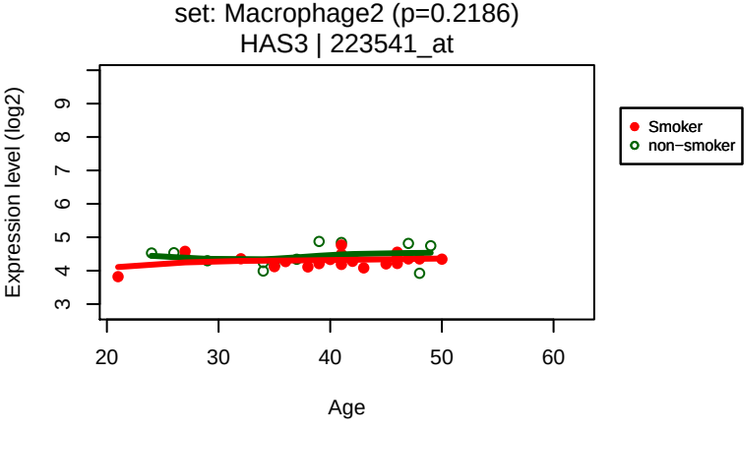

set: SmallAirway2 (p=0.983)
HAS3 | 223541_at
Expression level (log2)
Age
Smoker
non−smoker

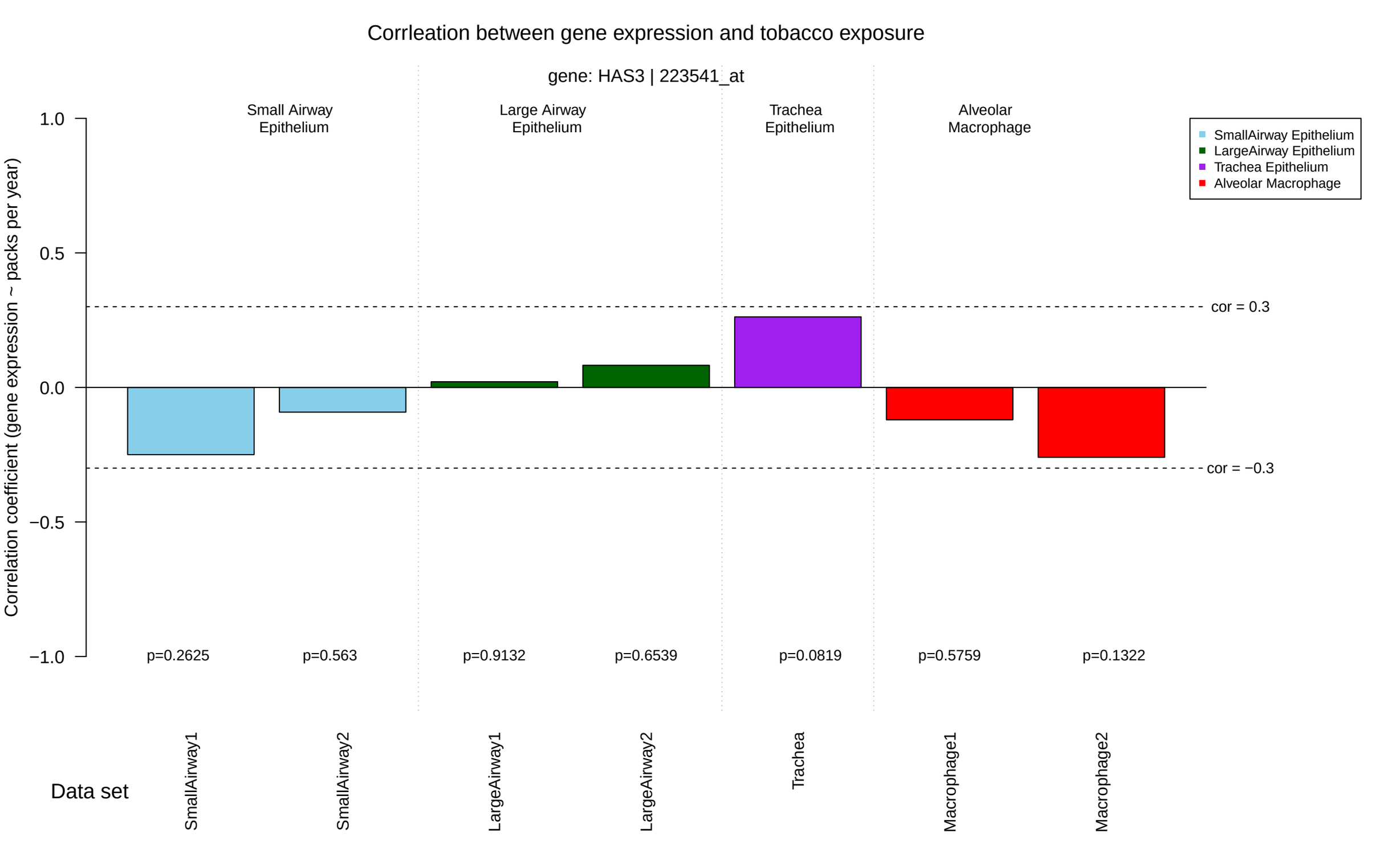

set: LargeAirway1 (p=0.7111)
HAS3 | 223541_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4039)
HAS3 | 223541_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.052)
HAS3 | 223541_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9113)
HAS3 | 223541_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2186)
HAS3 | 223541_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HAS3 | 223541_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2625
p=0.563
p=0.9132
p=0.6539
p=0.0819
p=0.5759
p=0.1322
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2